

#### What is Atherosclerosis:

Atherosclerosis is a disease in which plaque builds up inside your arteries. Arteries are blood vessels that carry oxygen-rich blood to your heart and other parts of your body. Plaque is made up of fat, cholesterol, calcium, and other substances found in the blood. Over time, plaque hardens and narrows your arteries, limiting the flow of oxygen-rich blood to your organs and tissues.

#### Etiology and Vascular Progression:

Atherosclerosis begins when the inner lining of an artery (the endothelium) becomes damaged. Common causes of endothelial damage include:

- High blood levels of low-density lipoprotein (LDL) cholesterol
- Elevated systemic blood pressure
- Chemical toxins from tobacco smoke
- Insulin resistance and uncontrolled diabetes
- Chronic systemic inflammation

Once damage occurs, platelets, cholesterol, and inflammatory cells adhere to the vessel wall, gradually creating a fibrous atherosclerotic plaque.

#### Associated Clinical Conditions:

Atherosclerosis can affect any artery in the body, leading to different clinical diseases:

- Coronary Artery Disease (CAD): When plaque narrows the coronary arteries supplying the heart muscle, leading to angina (chest pain) or heart attack.
- Carotid Artery Disease: When plaque builds up in arteries on the side of the neck supplying the brain, leading to stroke or transient ischemic attack (TIA).
- Peripheral Artery Disease (PAD): Plaque in arteries supplying blood to the limbs (typically the legs), causing claudication, pain, and tissue loss.
- Chronic Kidney Disease: Caused by narrowing of the renal arteries.

#### Diagnostic Testing and Clinical Management:

Diagnosis involves physical exams, blood lipid testing, echocardiograms, CT coronary angiograms, and stress testing. Management focuses on lifestyle modifications: adopting a Mediterranean or DASH dietary pattern, regular physical exercise, weight optimization, smoking cessation, and targeted medical management under a physician's care.